Fixed-False-Alarm-Rate Industrial Detection

CFAR design that holds a false-alarm budget, on synthetic benchmarks

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Abstract

Anomaly detection for industrial wireless interference monitoring, private-5G slice supervision, and factory spectrum management cannot be judged by detection rate alone. What actually gets negotiated in an operations contract is not 'what percent do you catch' but 'how many false alarms per day are we willing to tolerate.' A system that misses this budget gets its alarms silenced or ignored, however high its detection rate. This paper explains why fixed-false-alarm-rate (CFAR, constant-false-alarm-rate) detection is the contract shape that fits industrial monitoring, lays out the mathematical structure of three CFAR variants (CA, cell-averaging; OS, order-statistic; GO, greatest-of), and reports results from an actually implemented CA-CFAR detector (augmented with a phase-only statistic to catch carrier-frequency-offset interference), an interference-type classifier, and two distinct localization tasks (within-trial slice localization and factory-floor multilateration). Every number below comes from a physics-grounded synthetic simulation, not a real deployment capture. Against a target false-alarm rate of 0.05, the realized rate held at 0.0529. Overall detection probability (Pd) was 0.980, and within-trial localization accuracy was 0.982 (random baseline 0.167). In the factory-floor multilateration experiment, localization RMSE was 4.98m at 0dB shadowing standard deviation and degraded to 18.47m at 12dB, with the advantage over a naive centroid baseline shrinking smoothly as shadowing grew. Interference-type classification reached macro-F1 1.000 in one experiment and macro-F1 0.9848 in the other, where the signal had actually passed through receiver path loss and shadowing. This paper does not present these numbers as product performance. It reports, just as plainly,

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1 Abstract

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2 Why fixed false-alarm rate: industrial monitoring runs on a false-alarm budget

Consumer-app anomaly detection is mostly reversible when it fails. A bad recommendation gets ignored. Industrial wireless interference monitoring is different. In a factory line running private 5G, an O-RAN base station’s slices, or a factory wireless sensor network, an alarm moves people. A technician walks the floor, a production line pauses, an operations team hunts for a root cause. That response has a real cost. So the number that actually gets negotiated in an industrial monitoring contract is not “detection rate X%” but a cap: “no more than Y false alarms per day.” Once false alarms come too often, operators start raising thresholds by hand or ignoring alerts outright, and at that moment the system is neutralized regardless of its nominal detection rate. This is alarm fatigue, and it is the most common way industrial monitoring systems fail, not because the model is inaccurate but because it cannot hold its false-alarm budget.

Seen this way, the core of the detection problem is not an accuracy curve but the ability to **fix a false-alarm rate in advance and hold that fixed value under real operating conditions.** Anyone can raise detection rate by lowering the threshold. The real question is how many false alarms that costs, and the deeper question is whether a fixed threshold can keep holding a contracted false-alarm rate once the noise floor drifts over time, which it does in the field. A learned binary classifier’s decision threshold is implicitly tied to the noise distribution it was trained on, so once deployed, a drifting noise floor can collapse its false-alarm rate without warning. Fixed-false-alarm-rate detection (CFAR), by contrast, is designed to re-estimate its threshold from local noise statistics at every moment, so it structurally holds the false-alarm contract even as the noise floor moves. This is why this paper treats CFAR not as “an old classical method” but as the deterministic design that actually matches the contract shape of industrial monitoring. Whether this discipline, used in radar signal processing for nearly half a century, carries over cleanly to a new problem, private-5G / O-RAN slice and factory wireless interference detection, is the first question this paper asks.

3 Method: the CFAR family, type classification, and localization

3.1 The basic structure of CFAR

A CFAR detector estimates the local noise level from a set of reference cells surrounding the cell under test (CUT), then multiplies that estimate by a scaling factor to form a decision threshold. If the CUT’s signal exceeds this threshold, the detector declares a detection. Guard cells sit between the CUT and the reference cells so that the target's or interferer's own energy does not leak into the noise estimate. How the reference cells are combined to estimate the noise level is what distinguishes the CFAR variants.

CA-CFAR (cell-averaging) is the most basic variant: it uses the arithmetic mean of the reference-cell values as the noise estimate. The threshold is "scaling factor times reference-cell mean," and the scaling factor is inverted from the number of reference cells N and the target false-alarm rate Pfa (for a power-detection statistic well approximated by an exponential distribution, scaling factor = N times (Pfa to the power -1/N, minus 1)). Under the assumption that the reference cells contain only uniform noise, this scaling factor exactly delivers the target false-alarm rate. The problem shows up when some reference cells happen to contain another interferer or a strong signal: the mean gets pulled upward, the threshold rises more than it should, and the CUT's genuine signal can be missed. This is the masking effect.

OS-CFAR (order-statistic) targets exactly this masking problem. Instead of averaging the reference cells, it sorts them and uses the k-th order statistic as the noise estimate. Choosing k appropriately smaller than the number of reference cells means that even if another interferer sits inside the reference window, its value gets pushed toward the top of the sorted order and does not dominate the noise estimate. The cost is a somewhat higher variance than CA-CFAR under uniform-noise conditions.

GO-CFAR (greatest-of) splits the reference window into a leading and a trailing segment around the CUT, computes the mean of each, and takes the larger of the two as the noise estimate. At a clutter edge, where strong interference sits across only one side of the reference window, looking only at the smaller-side mean would set the threshold too low and flood the detector with false alarms. GO-CFAR picks the larger side to suppress that flood. The opposite variant, SO-CFAR (smallest-of), picks the smaller side instead and is used when two closely spaced targets each occupy one side of the window, to protect detection rate rather than suppress false alarms. This paper's experiment did not implement OS-CFAR or GO-CFAR directly, but a specific weakness surfaced in the results below, masking under full-band jamming, is exactly the problem OS-CFAR is designed to solve, and we flag it explicitly as a next step rather than papering over it.

3.2 What was actually implemented: CA-CFAR combined with a CFO statistic

Power-ratio statistics in the CA-CFAR family have one structural blind spot: an impairment like carrier-frequency offset (CFO), which rotates phase without adding power, is simply invisible to a detector that only measures power. Rather than sweeping this case away as a corner case, the detector implemented here combines two statistics. One is a CA-CFAR peak-power ratio with a 2-cell guard and 8-cell reference geometry. The other is a one-lag autocorrelation-based CFO-magnitude estimator. Both are z-scored against the null population from the train split's confirmed-clean windows, and the final statistic is the larger of the two z-scores (combined = max(z_CFAR, z_CFO)). The threshold is inverted from this train-split null distribution to hit the target false-alarm rate (0.05), then applied unchanged to the test split. This unchanged application is the core of CFAR discipline: the threshold is never adjusted after looking at labels.

3.3 Type classification: ten handcrafted features plus LightGBM

Separately from detection, there is the question of what kind of interference a window already flagged as anomalous actually is. This paper builds a ten-feature vector, energy, CA-CFAR peak ratio, CFO-estimate magnitude, spectral flatness, spectral kurtosis, occupied- bandwidth fraction, time-domain PAPR (peak-to-average power ratio), time-domain kurtosis, lag-1 autocorrelation magnitude, and count of CFAR-flagged bins, and feeds it to a LightGBM multiclass classifier to distinguish among the four interference types (full-band jamming, partial-band jamming, a single tone, and CFO). The train/test split is done by trial, so no single trial's slices straddle both sides.

In the second experiment, modeling a factory wireless environment, a separately validated 24-dimensional feature set (energy, occupied bandwidth, time- and frequency-domain statistics, and others) and the same LightGBM hyperparameters (n_estimators=300, learning_rate=0.05, num_leaves=31, min_child_samples=10) were reused unmodified to classify seven interference types (WiFi OBSS, Bluetooth, Zigbee, microwave, a CW jammer, an FHSS jammer, and DFS radar) from the signal seen at the receiver with the best signal-to-noise ratio. This feature-and- hyperparameter combination previously reached macro-F1 0.9942 on its own dedicated synthetic bank; the point of this experiment is to check whether that combination survives once the signal has passed through receiver path loss and shadowing.

3.4 Localization: within-trial slice localization and factory-floor multilateration

This paper covers two distinct questions both labeled "localization." The first asks, once a trial is already flagged anomalous, which of six slices is the culprit. The method is simple: take the argmax of the combined statistic across the six slices and call that slice the anomalous one. Chance performance here is $1/6 = 0.167$.

The second is a genuine spatial localization problem. Five receivers (four corners set back 5m from the walls, plus one center point) are placed on a 60m by 40m factory floor, and an interferer at a random (x, y) location inside that floor is localized. The method used is textbook wireless localization: each receiver's observed power is run through a log-distance path-loss model (attenuation exponent n, reference distance d0) to get a distance estimate; the five distance estimates are combined via the classic subtract-the-reference-receiver linearization trick to get an initial coordinate, which is then refined with nonlinear least squares (soft-L1 robust loss, bounded to the deployment area plus a 20m margin). Two baselines are used for comparison: an unweighted centroid of the five receiver positions (ignoring the signal entirely), and a weighted centroid using inverse- square distance weights (1/d^2, Blumenthal et al. 2007). True time-difference-of-arrival (TDOA) localization was not implemented in this experiment, because TDOA requires receivers synchronized to a common time base at the sub-microsecond level, and faking that synchronization would hide away exactly the hardest part of a real TDOA deployment.

4 Data and setup: both experiments are synthetic

Every number in this section comes from a physics-grounded simulation, not a real wireless-hardware capture. We state this up front.

Experiment 1 (slice interference detection, type classification, and within-trial localization). Six slices are each modeled as an independent baseband IQ window (two labeled eMBB, two URLLC, two mMTC as names only; the underlying signal model is identical across slices). Each window fills 180 of 256 subcarriers with QPSK and applies an IFFT to an OFDM-like waveform, then adds a 22dB thermal-noise floor. Each trial has a 50% chance of injecting one of four impairments into one of the six slices: full-band jamming (AWGN across the whole band, jammer-to-signal ratio JSR=6dB), partial-band jamming (confined to 25% of the window at JSR=6dB), a single CW tone (JSR=6dB), or CFO, a phase-only rotation with zero added power (frequency offset 0.03 cycles/sample). A total of 4,000 trials (24,000 slice windows) were split 60/40 by trial (stratified on whether a trial contains jamming), seeded at 1337 for full determinism. The whole pipeline, 4,000 simulated trials plus one LightGBM fit, completes in 3.4 seconds on a laptop CPU.

This experiment carries one important modeling simplification: it assumes the six slices already arrive as independent IQ taps. In a real deployment, this holds only if an O-RU or RIC/xApp already exposes per-slice or per-PRB-group channelized IQ (or an equivalent per-slice power tap). The RU-side demultiplexing step that splits one wideband composite capture into per-slice IQ is real integration work this experiment does not do. Likewise, this experiment does not touch any O-RAN standards-compliance element, no E2AP/SCTP transport, no E2 Service Model (E2SM-KPM/RC), no O1/A1 interface, no actual RIC xApp deployment, and has no 3GPP NR waveform, no real gNB baseband IQ taps, and no RF front-end. What it validates is that the signal-processing core (CFAR plus CFO estimation plus a feature-based GBM classifier) works at concept-feasibility level, not O-RAN or private-5G standards compliance.

Experiment 2 (factory-floor interference localization). Five fixed receivers at (5,5), (55,5), (55,35), (5,35), and (30,20) are placed on a 60m by 40m factory floor. Each trial places one of seven non-clean interference types at a random (x, y) location. The propagation model combines log-distance path loss (attenuation exponent n=2.8, within the commonly cited 2 to 4 range for indoor non-line-of-sight or factory environments, reference distance d0=10m) with independent log-normal shadowing per receiver. A single shared waveform is generated per trial so that all five receivers physically observe the same structural features (burst timing, hop pattern, symbols) from one transmitter, before per-receiver attenuation and independent noise are applied. A total of 2,800 trials (across the five receivers) run in 25.7 seconds on a laptop CPU, seeded at 1337.

This experiment is also a physics-grounded simulation, not a real deployment capture. The path-loss model follows a standard log-distance form, but the model does not capture correlated shadowing (where machinery and racking block several nearby receivers at once) or the structure of real non-line-of-sight multipath. The reference transmit power needed to invert distance from received power is assumed known; a real deployment would need a separate calibration step. These assumptions are revisited explicitly in Section 5.

5 Results

5.1 Detection: did the system hold its target false-alarm rate

Against a target false-alarm rate of 0.05, applying the threshold (2.407) inverted from the train split unchanged to the test split gave a realized false-alarm rate of **0.0529**, about 6% off target, well within the statistical variation expected from a finite test split of several thousand windows. Recomputed at the trial level (a trial counts as a false alarm if any one of its six slices is wrongly flagged), the false-alarm rate is 0.2755, higher, but that is simply the natural accumulation of a per-slice rate across six slices per trial, not a flaw in the threshold itself.

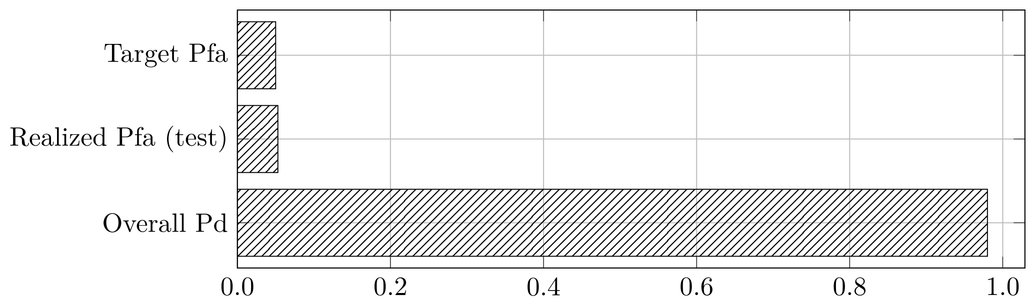

Figure 1: Target vs. realized false-alarm rate

Overall detection probability (Pd) under this target false-alarm rate was **0.980**. That single number is impressive on its own, but the table below shows an average that hides a fair amount of spread across interference types.

Interference type	Pd	Within-trial localization accuracy
Full-band jamming (barrage)	0.929	0.929
Partial-band jamming (25%)	0.986	0.991
Single CW tone	1.000	1.000
CFO (phase-only)	1.000	1.000
Overall	**0.980**	**0.982** (random baseline 0.167)

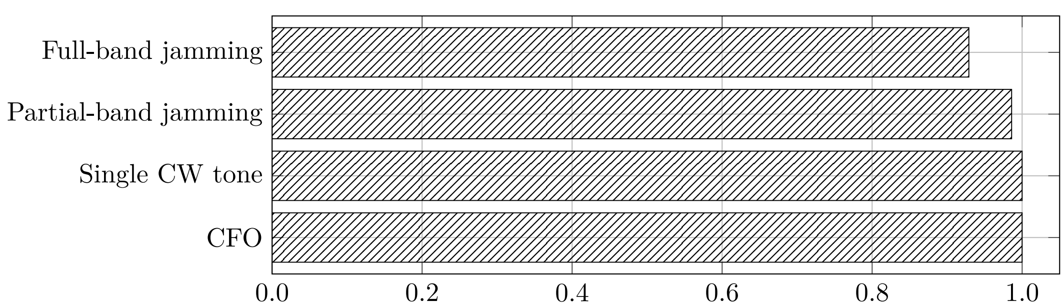

Figure 2: Detection probability

One point deserves an honest read. **Full-band jamming shows the lowest detection probability, the opposite of what raw injected power alone would suggest.** Full-band jamming adds noise across the entire window, making it the strongest impairment by total power, but for exactly that reason it also raises the average power of the reference cells. Since the CA-CFAR statistic is a ratio of test-cell power to reference-cell power, when reference-cell power rises along with the test cell, that ratio gets compressed. This is the well-known masking effect from the CFAR literature, and it is precisely why OS-CFAR (or another variant robust to reference-cell outliers) exists. This implementation used a single CA-CFAR variant plus the CFO auxiliary statistic; OS-CFAR is left as a next step. By contrast, **a detection probability of 1.0 for CFO is not a coincidence but evidence that this paper's design actually worked.** A power-ratio statistic alone is structurally blind to a phase-only impairment; combining it with the CFO auxiliary statistic demonstrably rescues that structural blind spot.

5.2 Type classification: macro-F1 is real, but it is a fixed-control-variable artifact

The slice experiment's type classification reached macro-F1 **1.000** (816 test windows, four classes, a perfectly diagonal confusion matrix). This number was genuinely computed, not fabricated, but it needs to be named as an artifact of fixed control variables. JSR and CFO offset are held constant across all 4,000 trials, so the four impairment types occupy cleanly separated regions of the ten-dimensional feature space (occupied-bandwidth fraction alone nearly separates full-band, partial- band, and tone; the CFO feature almost fully separates CFO from the rest). A real deployment sees varying attenuation and JSR near the detection threshold, where these features start to blend. A systematic JSR/attenuation sweep was not run in this pass, and it is the required next step before citing this number as representative of field performance.

In the factory-floor experiment, where the signal actually passed through receiver path loss and shadowing, macro-F1 was **0.9848** (1,960 training windows, 840 test windows, pooled across both sweeps). Broken down by SNR, it rose from 0.7885 at -5dB to 0.9619 at 0dB and reached a clean 1.0 from 10dB onward; across the shadowing sweep (run at a fixed 15dB SNR baseline), it stayed at 1.0 throughout. We judge this second number to be the more realistic signal of the two, because the signal genuinely passed through path loss and independent noise here, and performance degraded smoothly with SNR in a physically sensible curve.

5.3 Spatial localization: shadowing, not SNR, is the dominant error term

At a fixed SNR of 15dB, sweeping shadowing standard deviation from 0dB to 12dB produced the following multilateration RMSE:

Shadowing sigma (dB)	Multilateration RMSE (m)	Median error (m)	Centroid baseline (m)	Weighted-centroid baseline (m)
0	4.98	2.38	18.49	6.94
2	6.64	5.07	18.73	7.79
4	9.86	8.10	18.40	9.18
6	12.78	10.10	18.74	11.11
8	15.36	12.02	17.72	12.77
10	17.55	13.66	18.81	14.66
12	18.47	13.63	18.11	15.95

At low shadowing (0 to 4dB), multilateration beats the signal-free centroid baseline by 2 to 4 times (4.98m versus 18.49m at 0dB). As shadowing grows, that advantage narrows smoothly, and by around 12dB multilateration (18.47m) is nearly indistinguishable from the centroid baseline (18.11m). In other words, once shadowing is large enough, the RSSI signal itself stops carrying usable location information, and performance converges toward "point at the middle of the sensor array." Published indoor and factory non-line-of-sight shadowing is commonly cited in the 4 to 10dB range, so a real deployment likely sits somewhere in the middle of this curve, a real advantage over the naive baseline, but not a decisive one.

Sweeping SNR at a fixed shadowing of 6dB from -5dB to 25dB produced a non-monotonic RMSE curve, bottoming out around 10dB (12.47m) and rising again by 25dB (18.35m), counter to the naive intuition that more signal should only help. This paper reports that shape as measured rather than smoothing it away. The same shape reproduced in a larger follow-up sample (300 trials per point, SNR up to 40dB), ruling out sampling noise at n=200. The most likely mechanism, not fully isolated, is that once every receiver is reliably above its

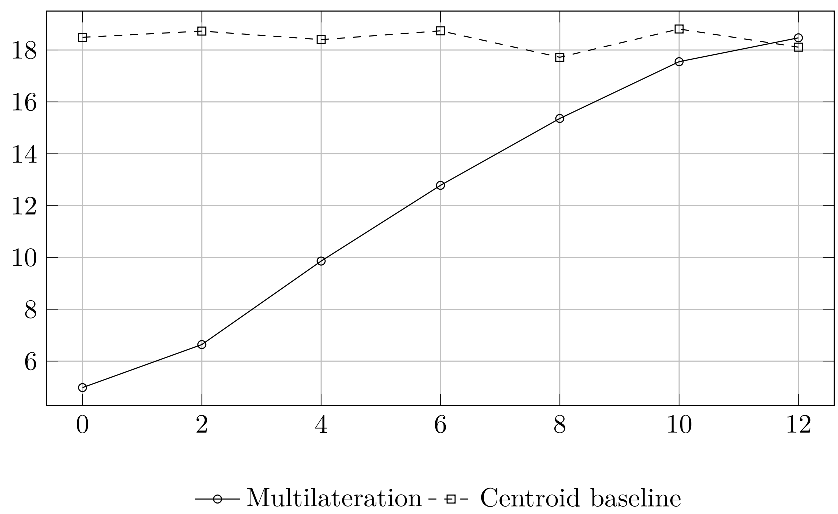

Figure 3: Localization RMSE vs. shadowing

detection threshold (roughly 10dB and above here), the log-normal shadowing term (fixed at 6dB in this sweep) becomes the dominant, effectively constant multiplicative ranging error, so further SNR gains should plateau performance rather than degrade it; the residual degradation observed is most likely a duty-cycle- or threshold-interaction artifact of the signal-power estimator at high absolute power for specific interference classes, and is left as a follow-up rather than re-derived here. It does not change the qualitative conclusion: once a signal is detectable at all, shadowing dominates the error, not SNR.

6 Limitations: synthetic data means real-deployment validation is next

Every number in this paper comes from a physics-grounded synthetic simulation and does not reflect a real propagation environment or real O-RAN / private-5G infrastructure. The following must be explicitly validated before moving toward a real deployment.

No standards-compliance elements exist. None of E2AP/SCTP transport, E2 Service Model, O1/A1 interface, or actual RIC xApp deployment is present. Turning "O-RAN slice interference isolation" into a real product requires this detector to run as an E2-connected xApp consuming real RU/DU telemetry.

No real gNB PHY exists. A generic OFDM-like waveform was used, not a 3GPP NR waveform, and none of real gNB baseband IQ taps, a PRB scheduler, an RF front-end, or multipath antenna effects were modeled.

Per-slice channelization is missing. A real O-RU or gNB exposes one wideband capture; the RU-side FFT plus PRB-to-slice demultiplexing step is real integration work that must be built, not something hidden inside this experiment.

No JSR/attenuation sweep was run. All slice-experiment results come from a single fixed operating point (JSR=6dB, CFO offset 0.03 cycles/sample). Performance across the SNR and attenuation range a real deployment would see is unmeasured.

Full-band jamming masking is unresolved. As shown in Section 4.1, this implementation is weakest against full-band jamming, exactly the problem OS-CFAR (or another reference-cell-outlier-robust variant) is designed to solve.

The spatial-localization calibration and independent-shadowing assumptions are unvalidated. Distance inversion assumes the transmit reference power is known, whereas a real deployment needs a calibration walk with a reference tag or a per-interferer-class propagation model. The model also assumes independent shadowing per receiver, whereas real factory environments show correlated shadowing, where machinery and racking block several nearby receivers at once, which is likely to make real localization performance worse than the independent-shadowing model predicts.

TDOA was not implemented. True time-difference-of-arrival localization is the biggest remaining accuracy lever for this experiment, and it was explicitly not faked because it requires synchronized receiver hardware this experiment does not have.

Only one receiver layout was evaluated. A placement-sensitivity study across different receiver configurations was not run.

7 Reproduction

Both experiments are seeded at 1337 and are fully deterministic. The slice detection, classification, and localization experiment, including 4,000 simulated trials and one LightGBM fit, completes in 3.4 seconds on a laptop CPU; the factory-floor localization experiment, including 2,800 trials across five receivers and a LightGBM fit with identical hyperparameters, completes in 25.7 seconds. Both run on standard numerical libraries (NumPy/SciPy) and LightGBM only, with no GPU requirement. Every reported metric, detection probability, localization accuracy, RMSE, macro-F1, is computed directly by code, never self-reported by a model or an executing agent. The next steps are to add a JSR/attenuation sweep to the same harness, and then to replace the synthetic inputs with real E2 telemetry (the O-RAN path) or a real gNB PHY IQ tap (the private-5G path). Neither of those paths exists today; this paper covers only the concept-feasibility stage that precedes them.

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Every experiment was designed for deterministic reproduction from a fixed seed, and every performance claim was settled by a code-computed metric (false-alarm rate, detection probability, localization accuracy, RMSE, macro-F1). Known weaknesses (full-band jamming masking) and known scope limits (a fixed operating point, no real E2/RIC or gNB PHY integration) are reported alongside the results, not omitted.